

Introduction to Azure DevOps

- Azure DevOps is Microsoft's comprehensive suite of development tools that supports the entire software development lifecycle (SDLC).
- Think of it as a complete digital factory for building software - from planning to deployment.
- Azure DevOps is a cloud platform by Microsoft that provides an end-to-end toolchain for:
 - **Planning** (Azure Boards)
 - **Source Code** (Azure Repos)
 - **Build Automation** (Azure Pipelines)
 - **Testing** (Azure Test Plans)
 - **Release & Deployment** (Azure Release Pipelines)
 - **Artifacts Storage** (Azure Artifacts)
- It's mainly used in **CI/CD pipelines**, DevOps automation, and team collaboration.
- Core Value Proposition:
 - Unified Platform: All tools work together seamlessly
 - Flexible: Works with any language, platform, or cloud
 - Scalable: From small teams to enterprise organizations
 - Integrated: CI/CD, version control, testing, and planning in one place
- The Five Pillars:
- Azure Boards - Work tracking
- Azure Repos - Source control
- Azure Pipelines - CI/CD automation
- Azure Test Plans - Testing management
- Azure Artifacts - Package management

Azure Repos

- Azure Repos provides Git repositories (distributed) or TFVC (centralized) for source control. Most modern teams use Git.
- Azure Repos provides unlimited private Git repositories and Team Foundation Version Control (TFVC) for your source code.
- Azure Repos primarily uses Git
- Azure Repos provides:
 - ✓ Git repositories
 - ✓ Branch policies
 - ✓ Pull Requests (PRs)

- ✓ Code reviews
 - ✓ CI integration
- It works just like GitHub but is tightly integrated with Azure DevOps.

Azure Boards

- Azure Boards is your agile planning tool that helps teams plan work, track progress, and collaborate efficiently.
- Work Item Types:
 - Epics - Large bodies of work that span multiple sprints
 - Features - Deliverable functionality within an epic
 - User Stories - Individual requirements from the user's perspective
 - Tasks - Specific work items needed to complete a story
 - Bugs - Issues found in the software
- The **Kanban board** visualizes your workflow with columns like To Do, In Progress, and Done.

Azure Pipelines Basics

- Azure Pipelines automates building, testing, and deploying your code. This is where CI/CD becomes reality.
- **Continuous Integration (CI)**: Every time developers push code, a pipeline automatically builds and tests it. This catches integration issues early.
- Pipeline Components:
 - Trigger - What starts the pipeline (code push, pull request, schedule)
 - Agent - The machine that runs your pipeline (Microsoft-hosted or self-hosted)
 - Stages - Major divisions like Build, Test, Deploy
 - Jobs - Units of work that run on an agent
 - Steps - Individual tasks like running a script or copying files
- Modern pipelines use YAML files stored in your repository
- The visual designer provides a GUI for creating pipelines without writing YAML.

Azure Test Plans

- Azure Test Plans provides comprehensive testing solutions for manual and exploratory testing.
- Test Plans Structure:

- Test Plans - Container for your testing effort (e.g., "Release 2.0 Testing")
- Test Suites - Logical groupings of test cases (e.g., "Login Features")
- Test Cases - Individual tests with steps and validation criteria